

Machine Learning Report

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1 Introduction

Our problem is to study the flue gas emissions of carbon monoxide (CO) and nitric oxide (NOx) from a gas turbine.

The dataset available for this study contains 36,733 instances, each corresponding to one hour of operation. For every instance, 11 different sensor measurements have been recorded and then aggregated over one hour, either by computing the average value or by summing the measurements. These data were collected from a gas turbine located in the north-western region of Turkey, which provides a real industrial context and makes the study directly relevant to practical applications.

The main objective of this work is twofold. First, we aim to build predictive models capable of estimating the emissions of CO and NOx based on the available sensor data. Second, we seek to analyze and understand the relationships between the different measured variables and the resulting gas emissions. By doing so, we want to identify which parameters have the most significant influence on the evolution of emissions over time and how variations in operating conditions affect the production of these pollutants.

From a mechanical engineering perspective, being able to predict and control gas emissions is essential for optimizing the performance and efficiency of gas turbines. Improving combustion efficiency, reducing pollutant formation, and meeting increasingly strict environmental regulations are all challenges that require a deep understanding of the combustion process and of the influence of operating conditions. The ability to manage gas releases in the most efficient possible way is therefore a key responsibility of a mechanical engineer, and this study contributes directly to developing the tools and knowledge needed to fulfill that role.

2 Data analysis

We first did an analysis on only 1 of the 5 datasets that we have, so we first selected a year, 2011.

First thing to know, is the data type and what we are actually measuring. So here is a table that sums that up :

Variable name	Abbreviation	Unit
Ambient temperature	AT	°C
Ambient pressure	AP	mbar
Ambient humidity	AH	%
Air filter difference pressure	AFDP	mbar
Gas turbine exhaust pressure	GTEP	mbar
Turbine inlet temperature	TIT	°C
Turbine outlet temperature	TAT	°C
Compressor discharge pressure	CDP	mbar
Turbine energy yield	TEY	MWh
Carbon monoxide	CO	mg/m ³
Nitrogen oxides	NOx	mg/m ³

Description of variables in the dataset

After identifying the variables, we examined their basic statistical properties to get a first sense of how each feature behaves. For every variable, we computed the main descriptive statistics such as count, mean, and standard deviation. Figure 1 illustrates this process using Ambient Temperature as an example. We observe that each feature contains over 7,400 entries (for one year), and key indicators such as the mean and standard deviation already provide an initial understanding of the data distribution.

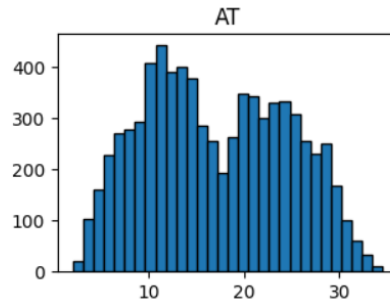
```

Statistiques
              AT
count  7411.000000
mean    17.112261
std      7.428307
min      2.116300
25%     10.992000
50%     16.366000
75%     23.344500
max     34.532000

```

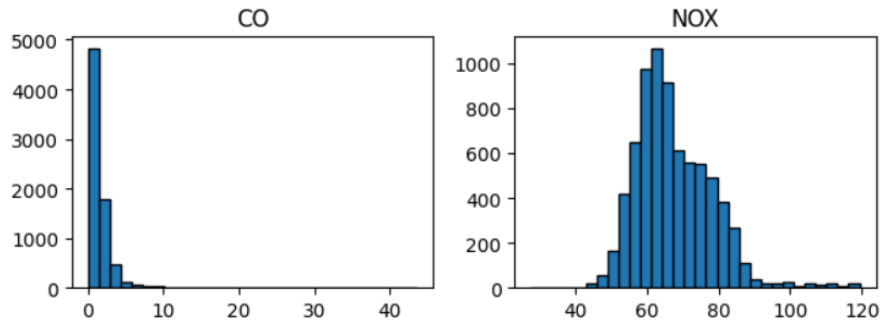
Ambient temperature statistical informations

To visualize these distributions clearly, we plotted histograms for each variable. The shape of the distribution helps identify potential outliers, or value range uniformity.



Ambient temperature data Visualization

As shown above for Ambient temperature, now let's see how our main variables evolve and are distributed :

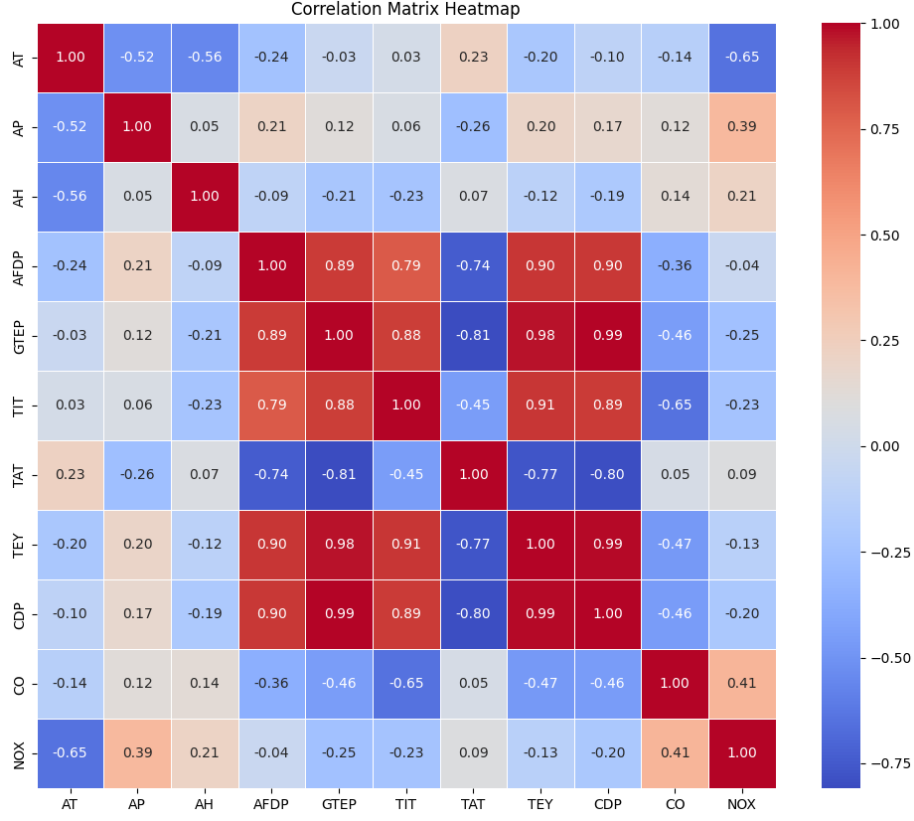


Ambient temperature data Visualization

As we can see, carbon monoxide (CO) is pretty well distributed around its mean, there are still some high values, There are some emissions peaks in the carbon monoxide (CO) emissions, and this will be really interesting to predict, maybe challenging.

In another hand, nitric oxide (NOx) is not well distributed around its mean and as also great values, there are also some emissions peaks to study.

We then moved on to studying the relationships between variables using a correlation heatmap.



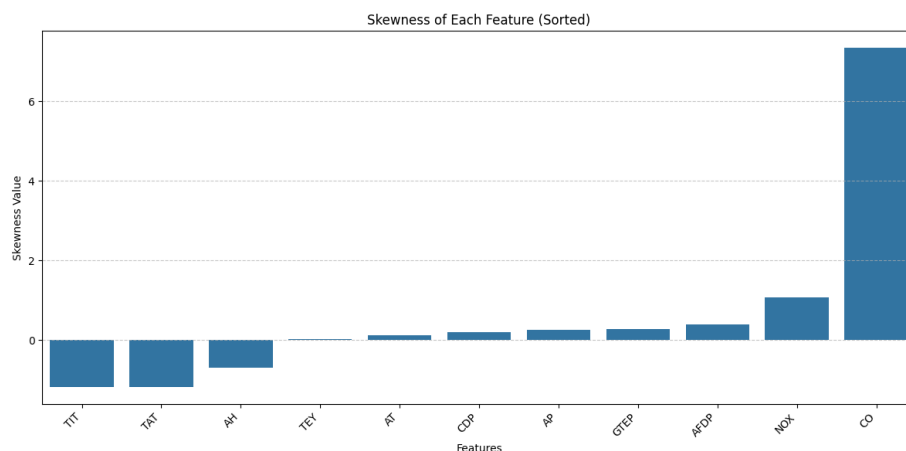
Heatmap

The heatmap requires a bit more interpretation. It works in such a way that values close to 1 indicate a strong positive correlation between two variables, while values close to 0 indicate weak or no correlation. Negative values represent inverse relationships, with values close to -1 meaning that when one variable increases, the other tends to decrease.

In practical terms, this correlation value reflects how well one variable can be approximated or explained by another. For example, the -0.81 correlation between TIT (Turbine Inlet Temperature) and TAT (Turbine Outlet Temperature) shows that these two temperatures evolve in opposite directions. Although not perfectly linear, variations indicate a strong dependency between the turbine's inlet and outlet thermal conditions.

In the context of our study, the heatmap reveals numerous strong correlations between several variables. This is why dimensionality reduction will be relevant in the data-refining phase, in order to reduce redundancy and keep only the most informative features.

Finally, we conducted a skewness analysis to quantify how asymmetrically each variable is distributed.



Skewness of each feature

Higher is the skewness, lower is the link between variables. Unlike the heatmap, which studies pairwise correlations, skewness evaluates each variable in relation to the entire dataset.

A key result is that carbon monoxide (CO) has a skewness close to 8 before refinement. This is problematic because such an unbalanced distribution makes CO harder to predict reliably from the other variables.

To complete the preliminary assessment, we also checked for missing values, duplicates, and potential type inconsistencies (such as text in numerical columns).

```
Missing values :
AT      0
AP      0
AH      0
AFDP    0
GTEP    0
TIT     0
TAT     0
TEY     0
CDP     0
CO      0
NOX     0
dtype: int64
```

```
Duplicates number : 0
```

Missing values and Duplicates count

As shown in Figure 5, the dataset does not contain missing entries or duplicate rows. This confirms that the dataset is clean and suitable for machine learning applications without requiring major preprocessing.

Summary : The dataset presents no structural quality issues (no missing values, no duplicates, consistent measurement types).

The correlation and skewness analyses suggest that predicting nitrogen oxides (NOx) should be relatively straightforward, whereas predicting carbon monoxide (CO) will be more challenging due to its highly skewed distribution and weaker alignment with other variables.

3 Data refining

The goal of this part was to fix as much as we could the dataset to make it good for the models we wanted to implement afterwards.

We began by implementing three algorithms which goal were respectively to fix the missing values, the inconsistencies and the duplicates in the dataset.

As said earlier, we had any problem with the dataset of year 2011 in terms of missing values and duplicates.

However, We didn't know if mixing all the 5 years together (final objective) or just another dataset like for example the 2015 one would pose problems.

That's why those algorithms needed to be implemented.

```

--- Verification and management of data inconsistencies ---
Number of missing values : 0
No data type incoherence.

NO incoherence or missing value in the dataset.No action needed.
Data types after verification and management of data inconsistencies :
AT      float64
AP      float64
AH      float64
AFDP    float64
GTEP    float64
TIT     float64
TAT     float64
TEY     float64
CDP     float64
CO      float64
NOX     float64
dtype: object

```

Missing values and Inconsistencies management

As we see here, there is no inconsistencies as well in the dataset. Inconsistencies appear when there are some values that are not meant to be there or in a different format than the one we want.

So here the inconsistencies program erases the inconsistencies by setting it to null and then the missing values program will drop those afterwards.

After that, we began the models implement preparation, so we divided the dataset into a training and a testing part. In this split we did one for carbon monoxide (CO) and another one for nitric oxide (NOx). Indeed, we wanted to analyze those variables, so put them into the training part would have been unproductive.

```
• --- NOX Model Split & Scaling ---  
  Shape of X_train_NOX: (5928, 9)  
  Shape of X_test_NOX: (1483, 9)  
  Shape of y_train_NOX: (5928,)  
  Shape of y_test_NOX: (1483,)  
  
  --- CO Model Split & Scaling ---  
  Shape of X_train_CO: (5928, 9)  
  Shape of X_test_CO: (1483, 9)  
  Shape of y_train_CO: (5928,)  
  Shape of y_test_CO: (1483,)
```

Splitting and scaling

In the capture of the code, we wrote for title "splitting and scaling" because there is also a scaling step in this algorithm.

Scaling is a very important part in Machine learning because it allows the data to be in a same equality state, so any data, with higher units would not have a bigger part in the analysis than others.

So we did transform the dataset by scaling, but only the one that have already been divided into two parts.

It is important because it ensures the conservation of data through the whole process.

4 Model implementation

The models we selected (Ridge, Lasso, Random Forest, Gradient Boosting, XG-Boost, and KNN) are relevant because they cover a broad range of modeling strategies.

This diversity is important in our case, since CO and NOx emissions depend on several industrial variables whose relationships can be linear, nonlinear, strongly correlated, or noisy. Using multiple model families allows us to determine which one captures the underlying physical behavior of the gas turbine most accurately.

To make the models the more relevant possible, we had to choose the best hyperparameters for them (the other parameters are defined in a classic way). That's why we created a pipeline for all models, to check if there was no data leakage by too late scaling or partial hyperparameters implementation.

Here we made a GridSearchCV in order to do a 5 Fold cross validation for each hyperparameter testing (for each model)

4.0.1 R² and MSE

To evaluate the performance of our models, we rely on two complementary metrics:

R² (coefficient of determination) and MSE (Mean Squared Error).

R² measures how much of the variability in the target variable is explained by the model. A higher R² indicates that the model succeeds in capturing the underlying structure of the data, whereas a lower R² suggests that the relationship is either weak or highly nonlinear.

MSE, on the other hand, quantifies the average squared difference between the predicted and true values. It provides an absolute measure of the model's error, which is heavily influenced by the natural spread of the target variable.

A variable with a wide and dispersed distribution, like NOx, typically leads to higher MSE values, while more compact variables, such as CO, naturally yield lower MSE. Together, R² and MSE give a balanced understanding of both relative explanatory power and absolute prediction accuracy.

4.1 Ridge Regression

Ridge is a linear regression model that adds an L2 penalty to reduce the impact of correlated features. Since many sensor variables in the turbine are strongly correlated (temperatures, pressures, etc.), Ridge stabilizes the coefficients and avoids extreme weight values. This makes it a useful baseline model for understanding the global linear tendencies in CO and NOx emissions.


```

Ridge for NOx
Fitting 5 folds for each of 3 candidates, totalling 15 fits
Best params: {'model__alpha': 0.1}
Test R2: 0.6178 | MSE: 6.3569

```

```

Ridge for CO
Fitting 5 folds for each of 3 candidates, totalling 15 fits
Best params: {'model__alpha': 0.1}
Test R2: 0.5049 | MSE: 1.1810

```

The Ridge model shows moderate performance, with an R^2 of 0.62 for NOx and 0.50 for CO. These values indicate that the model captures part of the linear structure in the data but cannot fully explain the variability of the emissions, which is expected given that turbine behavior often involves nonlinear interactions between temperature, pressure, and flow conditions.

4.2 Lasso Regression

Lasso performs linear regression with an L1 penalty, which forces some coefficients to become exactly zero. Concretely, it selects only the most relevant variables while discarding the least informative ones. This is valuable in our dataset because several input features convey redundant information. Lasso helps identify which sensor measurements have the strongest influence on emissions

```

Lasso for NOx
Fitting 5 folds for each of 3 candidates, totalling 15 fits
Best params: {'model__alpha': 0.001, 'model__max_iter': 20000}
Test R2: 0.6180 | MSE: 6.3549

```

```

Lasso for CO
Fitting 5 folds for each of 3 candidates, totalling 15 fits
Best params: {'model__alpha': 0.001, 'model__max_iter': 20000}
Test R2: 0.5072 | MSE: 1.1781

```

Lasso gives results very similar to Ridge for both NOx and CO, with nearly identical R^2 and MSE values. This is expected because Lasso performs linear modeling while applying L1 regularization, which mainly affects coefficients when some features are redundant or weakly influential.

4.3 Random Forest

Random Forest builds many decision trees using random subsets of features and samples, then averages their predictions. This allows it to capture nonlinear relationships, interactions between variables, and sudden changes in operating conditions that linear models cannot model. It is particularly suited to turbine emissions, which depend on complex thermodynamic interactions.

```
RandomForest for NOx
Fitting 5 folds for each of 4 candidates, totalling 20 fits
Best params: {'model__max_features': 'sqrt', 'model__n_estimators': 200}
Test R2: 0.9110 | MSE: 3.0677
```

```
RandomForest for CO
Fitting 5 folds for each of 4 candidates, totalling 20 fits
Best params: {'model__max_features': 'sqrt', 'model__n_estimators': 200}
Test R2: 0.6316 | MSE: 1.0187
```

Random Forest significantly improves performance compared to linear models, reaching an R^2 of 0.91 for NOx and 0.63 for CO. This indicates that nonlinear interactions between variables—such as combined effects of temperature, pressure, and humidity—play a major role in emission levels.

4.4 Gradient Boosting

Gradient Boosting constructs trees sequentially, with each new tree correcting the mistakes of the previous ones. This makes it effective for capturing subtle patterns and rare events in the data. Since CO and NOx both exhibit irregular variations—especially CO—Gradient Boosting can model small but meaningful deviations in operating behavior that simpler models miss.

```
GradientBoosting for NOx
Fitting 5 folds for each of 8 candidates, totalling 40 fits
Best params: {'model__learning_rate': 0.1, 'model__max_depth': 4, 'model__n_estimators': 200}
Test R2: 0.8954 | MSE: 3.3257
```

```
GradientBoosting for CO
Fitting 5 folds for each of 8 candidates, totalling 40 fits
Best params: {'model__learning_rate': 0.1, 'model__max_depth': 3, 'model__n_estimators': 200}
Test R2: 0.5159 | MSE: 1.1677
```

Gradient Boosting also performs very well on NOx with an R^2 close to 0.90, demonstrating its ability to capture complex emission dynamics. The slightly higher MSE compared to Random Forest reflects the model's sensitivity to noise, but overall performance remains strong.

4.5 XGBoost

XGBoost is an optimized version of gradient boosting with improved regularization and computational efficiency. It handles noise, outliers, and complex nonlinearities extremely well. Given that turbine emissions are influenced by multiple interacting factors, XGBoost is a strong candidate for accurate predictions, especially when the relationships are not smooth or uniform.

```
XGBoost for NOx
Fitting 5 folds for each of 8 candidates, totalling 40 fits
Best params: {'model__learning_rate': 0.1, 'model__max_depth': 5, 'model__n_estimators': 200}
Test R2: 0.9067 | MSE: 3.1413
```

```

XGBoost for CO
Fitting 5 folds for each of 8 candidates, totalling 40 fits
Best params: {'model__learning_rate': 0.1, 'model__max_depth': 3, 'model__n_estimators': 200}
Test R2: 0.5907 | MSE: 1.0737

```

XGBoost gives results close to Gradient Boosting but generally more stable, with an R^2 above 0.90 for NOx and around 0.59 for CO. This confirms XGBoost’s strength in handling complex nonlinear relationships while mitigating overfitting through built-in regularization.

4.6 K-Nearest Neighbors (KNN)

KNN predicts a value by averaging the outputs of the closest samples in the dataset. In practice, this means it captures local behavior: similar operating conditions lead to similar emissions. This is highly relevant for our turbine data, where emissions depend on local clusters of sensor readings and small variations in temperature or pressure. KNN performed best because NOx and CO emissions follow strong local patterns that this method naturally models.

```

KNN for NOX
Fitting 5 folds for each of 6 candidates, totalling 30 fits
Best params: {'model__n_neighbors': 3, 'model__weights': 'distance'}
Test R2: 0.9174 | MSE: 2.9551

```

```

KNN for CO
Fitting 5 folds for each of 6 candidates, totalling 30 fits
Best params: {'model__n_neighbors': 7, 'model__weights': 'uniform'}
Test R2: 0.6428 | MSE: 1.0031

```

KNN surprisingly performs the best for NOx, achieving an R^2 of 0.917 and the lowest MSE among all models. This indicates that NOx behaves in a highly locally predictable way: instances with similar sensor readings yield similar emission levels.

The chosen hyperparameters (3 neighbours and distance weighting) reinforce this local structure. For CO, KNN also performs well with the best MSE and an R^2 of 0.64, suggesting that CO emissions exhibit strong neighbourhood patterns despite having lower overall variance. Since KNN makes no assumptions about linearity or global structure, it adapts naturally to locally clustered industrial data

Comparison for NOX		
Model	R2	RMSE
KNN	0.9174	2.9551
RandomForest	0.9110	3.0677
XGBoost	0.9067	3.1413
GradientBoosting	0.8954	3.3257
Lasso	0.6180	6.3549
Ridge	0.6178	6.3569
Comparison for CO		
Model	R2	RMSE
KNN	0.6428	1.0031
RandomForest	0.6316	1.0187
XGBoost	0.5907	1.0737
GradientBoosting	0.5159	1.1677
Lasso	0.5072	1.1781
Ridge	0.5049	1.1810

Summary : By combining linear models, nonlinear tree-based methods, boosting algorithms, and a non-parametric approach, we obtain a comprehensive view of the dataset.

This allows us to compare performance across different modeling philosophies, understand which variables influence emissions, identify nonlinearities, and ultimately determine the most suitable model for predicting CO and NOx.

5 Adjustments and improvements

As we just saw, the KNN is the best model to approximate and predict the carbon monoxide (CO) and nitric oxide (NOx) behavior, now we will do tests and technics to make it better.

5.1 Dimension reduction

Firstly, the Dimension reduction, so instead of just testing the model on a testing sample with all the variable and data on it.

We decided to do a dimension reduction to upgrade the speed, efficiency and stability of the model.

Applying PCA in this context turned out to be counter-productive. PCA reduces dimensionality by preserving global variance, but it does so at the cost of distorting the local geometry of the data.

Metric	NOX (No PCA)	NOX (With PCA)	CO (No PCA)	CO (With PCA)
R2	0.917397420507559	0.8971829306587054	0.6427912251982966	0.635704447968529
RMSE	2.9550992323251717	3.296915943033456	1.003068708378022	1.0120669389452802

In a lot of models it is interesting, but, since KNN relies entirely on meaningful distances and neighborhood structures, altering these distances leads to weaker predictions.

Important fine-scale patterns that help identify similar operating conditions are lost in the compressed PCA space.

As a result, performance decreases without bringing any real benefit in speed or stability for our dataset.

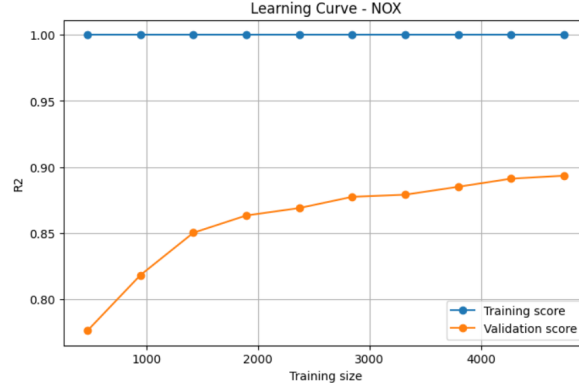
5.2 Overfitting - Underfitting ?

This part is also really important to check, indeed it shows us how the model will work with new datasets, and be able to predict effectively.

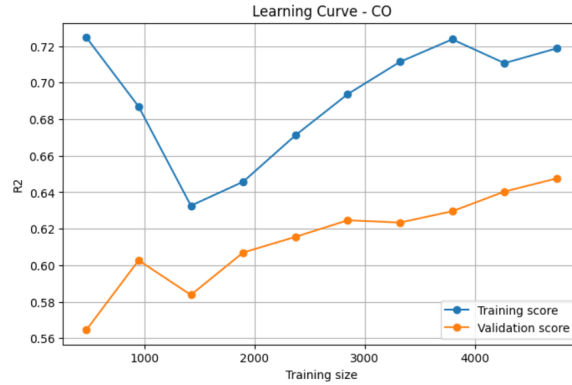
```
=== Diagnostic for Best NOX Model ===  
Train R2: 1.0000  
Test R2 : 0.9174  
Balanced model (no strong overfitting).
```

```
=== Diagnostic for Best CO Model ===  
Train R2: 0.7249  
Test R2 : 0.6428  
Balanced model (no strong overfitting).
```

In here we can see, our model have some losses in practical area compared to only theory. Though we can see a better representation of the problem by tracing learning curves for the two variables.



For NOx, the training score remains close to 1.0 while the validation score steadily increases with the training size, approaching 0.90. This pattern indicates that the model benefits significantly from additional data and generalizes well, suggesting that NOx emissions follow stable and learnable relationships with the sensor variables.



In contrast, CO presents a much noisier learning curve: both training and validation scores fluctuate at smaller sample sizes, and the validation score increases only slowly, reaching around 0.64. This reflects the intrinsic complexity of CO, whose distribution is highly skewed and less correlated with the other features. As a result, the model cannot fully capture its underlying structure, even with more training data.

Overall, the curves confirm that NOx is substantially easier to predict, while CO exhibits limited learnability due to its irregular and volatile behavior.

5.3 Model Stacking

A stacking regressor should improve results because it combines the strengths of several different models instead of relying on just one.

Each model—linear, tree-based, distance-based, or boosting—captures different patterns in the data. Some models are better at modeling smooth global trends, others identify local behaviors, and others detect nonlinear interactions.

By stacking them, the meta-model (often a linear regressor or another tree-based model) learns how to weight and combine their predictions in the most effective way.

	Metric	NOX (KNN)	NOX (Stacking)	CO (KNN)	CO (Stacking)
0	R2	0.917397	0.926194	0.642791	0.620839
1	RMSE	2.955099	2.793328	1.003069	1.033432

The results here are means that the KNN is not helped at all by other prediction models.

Indeed, all the other models prediction are included in the KNN one. This is though really bad for our study because with our actual tools we can't move forward in optimization

6 Conclusion

In this project, we investigated the behavior of CO and NOx emissions from a gas turbine using a variety of machine learning models. Through detailed exploratory analysis, we established that the dataset is clean and structurally reliable, but that the two target variables present very different levels of predictability. NOx is relatively well-behaved, correlated with several operating variables, and follows stable patterns, whereas CO is highly skewed, more irregular, and only loosely related to the other measurements. This fundamental difference strongly influenced the model performances.

Among all the tested methods, K-Nearest Neighbors (KNN) emerged as the best-performing model, achieving the highest R^2 and lowest error for both pollutants, especially NOx. This result confirms that emissions in this turbine exhibit strong local structures that KNN is uniquely suited to capture. Tree-based models such as Random Forest or XGBoost also performed well, particularly on NOx, showing that nonlinear interactions play a significant role in emission dynamics. Linear models, on the other hand, were too limited to capture the complexity of the system.

Additional techniques such as PCA dimensionality reduction and stacking were explored to further improve performance. However, PCA degraded the results because it distorts local distances—an essential component for KNN. Likewise, stacking did not bring improvements, indicating that the predictive power of the auxiliary models is already implicitly embedded in KNN’s behavior and cannot meaningfully complement it.

Several avenues remain open for improvement. A deeper treatment of CO’s skewness—such as log-transforming the variable, applying quantile regression, or using models designed for heavy-tailed distributions—may enhance its predictability. More advanced ensemble methods, including CatBoost or hybrid local-global models, could further refine performance. Finally, expanding the dataset to include all available years, and analyzing temporal dependencies, might reveal additional structure not captured in this study.

Overall, this project demonstrates the importance of combining data analysis, model diversity, and diagnostic tools to understand and predict industrial emissions. The insights gained here contribute both to machine learning methodology and to the operational understanding of gas turbine behavior.